

The diagram illustrates an identity for the imaginary part of a self-energy diagram. On the left, a square loop with a vertical line on the right is connected to a vertex. From this vertex, a horizontal line with an arrow pointing right is labeled Im . This is followed by an equals sign. On the right side of the equation, there are two terms separated by a plus sign. The first term is a square loop with a vertical line on the right, connected to a vertex, which then connects to a horizontal line with an arrow pointing right. The second term is a horizontal line with an arrow pointing right, labeled Im above it and 0 to its right.

$$\text{Diagram 1} \xrightarrow{\text{Im}} = \text{Diagram 2} + \text{Diagram 3}$$